

Lab 02 - First Verilog

In this lab, you've learned how to do an initial and simple design in Verilog to learn the Vivado tooling and process involved in RTL/FPGA design.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

Summarize your learnings from the lab here.

In this lab, we built a Verilog circuit that makes a switch flicking turn an LED on or off. It taught us how to work in Vivado and get control over circuit boards. The steps felt very overwhelming at first but now that we understand them it will be much easier next time to program a circuit board.

Lab Questions

1 - Describe the stages of building a Verilog project in Vivado.

Go to Vivado -> Create Project, name your project and give it a directory, specify the project type as RTL, add files, customize files, add constraints, select a board, review the summary. Then, write RTL (the code), simulate it, run synthesis, run the implementation, make a bitstream, and program it onto the device.

Or in a more simplified way, set up the project, write RTL (instructions), practice (simulate), synthesis (build), implement (locate then wire), bitstream (compile), program (load).

2 - What is the value in looking at the elaborated design schematic?

The value in looking at the elaborated design schematic is because it provides a detailed version of the product, allowing you to see how everything will actually work before building it. It's like a blueprint showing what modules connect to what before actually assembling them together.

3 - Why should we simulate our designs frequently? What does the simulation do?

We simulate our designs frequently because it's a bit like building out your blueprint digitally before actually assembling it. Imagine if you designed a bridge it's a lot faster to see if it'll work via 3D models and simulations than trying to build the full thing in real life and testing to see if it'll work. In other words, it lets us see if the design is going to work in a cheap, fast way.

Simulation makes the Verilog code run against test inputs and lets us see what every signal is going to do. So it lets us see mistakes before they're expensive.

Code Submission

Upload a .zip of all your code or a public repository on GitHub.

<https://github.com/Haydennn350/ECE-230L-Lab-2.git>